

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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Motivation: Ownership Changes Beyond Market Structure

- Ownership transitions are common in healthcare.
- IO research often emphasizes **prices, market power, and concentration**.
- But many transitions do **not** change the set of providers in a market.
- In those cases, effects operate through **internal production decisions**: management practices and especially **staffing inputs**.

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
- Do responses differ across staff type or organizational structure?
- Do responses differ pre vs post pandemic?
- Why we care: staffing is the primary input into care, so ownership changes can affect **quality** even when market structure is unchanged.

Why Nursing Homes?

- Nursing home care is labor-intensive and heavily regulated.
- **Nursing labor is the main input** into care and a major operating cost.
- Staffing is a natural adjustment margin after ownership change, but it's constrained by:
 - ▶ compliance,
 - ▶ resident outcomes,
 - ▶ demand/occupancy

Nursing Staff Types

- We study four staffing measures:
 - ▶ **RN (Registered Nurses):** clinical assessment, care planning, meds, supervision
 - ▶ **LPN (Licensed Practical Nurses):** routine clinical tasks, meds under RN oversight
 - ▶ **CNA (Certified Nursing Assistants):** direct daily care
 - ▶ **Total nursing:** RN + LPN + CNA
- Ownership change can affect **levels** (total hours) and **mix** (skill composition).

The Cost–Quality Tradeoff in Staffing

- Owners can raise quality through:
 - ▶ **Quantity** of staffing and/or
 - ▶ **Quality / skill mix**
- Either margin increases costs
- But **revenues are relatively flat in the short run**
- So owners face a tradeoff: **higher quality vs. tighter margins.**

- **Cost pressures:** tighter budgets / debt service can push cost cutting.
- **Reorganization:** standardized policies may shift RN/LPN/CNA composition.
- **Quality objective:** owners with stronger quality/compliance goals may increase staffing.
- Net effect is ambiguous $\Rightarrow$ **empirical question.**

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 - ▶ **Initial flag** that a change may have occurred

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 - ▶ Monthly ownership panel with every reported ownership layer (managing org → direct owners → indirect owners)
 - ▶ Used to **identify** and **date** ownership changes using ownership shares

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 - ▶ Payroll-reported staffing hours by staff type
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- facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024

Identifying Ownership Changes: Why Ownership Is Non-Trivial

- Ownership is often **layered** (organizations owning organizations)
- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Entity Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes: Example (Legal Restructuring)

- HCRIS ownership change flags may reflect **purely legal restructuring**
- NHC ownership data allows us to verify whether economic control changes
- **Example from the data:**
 - ▶ *Before:*
 - ★ ESTES JAMES (84%)
 - ★ BURCHFIELD JOHN (10%)
 - ★ LEE CLAUDE (6%)
 - ▶ *After:*
 - ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
 - ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- We do **not** treat this as a meaningful ownership change

Identifying Ownership Changes: Definition and Sample

- **Step 1 (flag):** HCRIS ownership change flag identifies candidate change events
- **Step 2 (validate & date):** NHC ownership panel identifies whether control changes and when
- **Ownership change criteria (NHC):**
 - ▶ An ownership change occurs between $t - 1$ and t when **majority control (50%)** transfers
 - ▶ Partial / missing shares are normalized; missing ownership shares are assigned equal weights within a level
 - ▶ We do not flag changes driven by name changes or within-family reclassification (e.g., trust/LLC relabeling)
- **Final sample restriction):**
 - ▶ Facilities with **(0,0)** changes in both datasets, or **(1,1)** with dates within 6 months
 - ▶ Change month is the first month NHC indicates a change

Ownership Change Agreement: NHC vs HCRIS

Cross-tab of Ownership Changes: NHC vs HCRIS

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)
		0	1	2+
		Count of changes in HCRIS		

Dependent Variables: Nurse Staffing (PBJ)

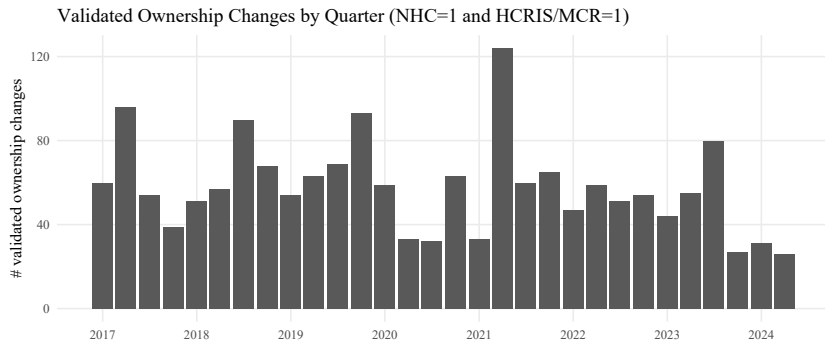
- We study four staffing measures:
 - ▶ **Registered Nurses (RN)**: clinical assessment, medication administration, supervision
 - ▶ **Licensed Practical Nurses (LPN)**: clinical support, medication-related tasks
 - ▶ **Certified Nursing Assistants (CNA)**: direct daily care (bathing, feeding, mobility)
 - ▶ **Total nursing staff**: RN + LPN + CNA
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

PBJ Cleaning and Final Sample

- We apply CMS technical guidance to remove implausible staffing reports:
 - ① RN HPPD = 0 **and** LPN HPPD = 0
 - ② Total HPPD < 1.5 or > 12
 - ③ CNA HPPD > 5.25
- Additional restrictions:
 - ▶ Drop facilities with beds < 15
 - ▶ Drop hospital-based facilities
- **Final sample:**
 - ▶ 7,806 facilities
 - ▶ 632,231 facility-month observations
 - ▶ 1,589 facilities undergo a change of ownership, about 20% of the final sample

Validated Ownership Changes Over Time (Quarterly)



Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD event study around ownership transitions
- Identification comes from **within-facility changes** relative to facilities not yet treated

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (1)$$

- Y_{it} : staffing outcome measured in **HPPD**
- α_i facility FE, λ_t month FE; X'_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- Standard errors clustered at the facility level

Identification Assumption

- **Parallel trends:** absent an ownership change, treated facilities would have followed the same staffing trends as facilities not yet treated.
- **Primary threat: anticipation / timing mismatch**
 - ▶ Staffing may adjust before the recorded transition month.

Why Anticipation is Plausible in Ownership Transitions

- Ownership changes are not instantaneous: negotiations and operational handoffs occur before the recorded change month
- Staffing can adjust during transition planning (turnover, hiring freezes, contract renegotiations)
- Administrative “event dates” may reflect reporting/legal dates, not the first operational change
- **Implication:** months immediately before $\tau = 0$ may violate parallel trends

Main Specification: Donut Event Study

- To handle this, we exclude months $\tau \in \{-3, -2, -1\}$
- We use $\tau = -4$ as the reference period
- Post change coefficients compare to a pre-period less likely to be influenced by transition activity

Assessing Parallel Trends

- We test parallel trends using a joint Wald test of

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- **Takeaway:**

- ▶ With the full pre-window (including $\tau = -3, -2, -1$), pre-trends show evidence of violation.
- ▶ Excluding anticipatory months (donut) yields pre-trends more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	51.22 (23) [0.0006]	49.22 (23) [0.0012]
2 Year Window with Donut	24.92 (20) [0.2046]	22.26 (20) [0.3268]
1 Year Window with Donut	6.08 (8) [0.6387]	6.33 (8) [0.6102]

Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report χ^2 (df) [p-value].

Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

Full outcome-by-staff-type table available in appendix.

Summary Effect: TWFE Difference-in-Differences

$$Y_{it} = \beta Post_{it} + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- $Post_{it} = 0$ before an ownership change and $= 1$ in all months after a change
- β summarizes the average post-change staffing effect
- Quantifies average changes for cleaner and relevant results